

## 1. Distinctions

| Concept                                                   | Explanation                                                                                                                                                          |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stochastic Process vs Filtration                          | A stochastic process $\{X_t\}$ is a collection of random variables indexed by time; a filtration $\{\mathcal{F}_t\}$ is the accumulated information up to time $t$ . |
| Super-martingale vs Sub-martingale                        | For a super-martingale, $E[X_{t+1} \mathcal{F}_t] \leq X_t$ ; for a sub-martingale, $E[X_{t+1} \mathcal{F}_t] \geq X_t$ .                                            |
| Single channel–single phase vs Single channel–multi phase | One server, one service stage vs one server, multiple sequential stages.                                                                                             |
| Multi channel–single phase vs Multi channel–multi phase   | Multiple parallel servers, one stage vs multiple servers, multiple stages.                                                                                           |
| $\sigma$ -algebra vs Hidden Markov Chain                  | A $\sigma$ -algebra is a mathematical structure defining measurable events; an HMM is a probabilistic model of hidden states and observable outputs.                 |
| Higher-order vs Time-homogeneous Markov chain             | Higher-order depends on multiple past states; time-homogeneous means transition probabilities are time-invariant.                                                    |
| Counting Process vs Branching Process                     | Counting process counts events; branching process models reproduction across generations.                                                                            |
| Communication class vs Absorbing class                    | Communication class: mutually reachable states; absorbing class: cannot be exited once entered.                                                                      |
| Transient vs Recurrent state                              | Transient: may be left forever; recurrent: revisited infinitely often.                                                                                               |
| Irreducible vs Reducible chain                            | Irreducible: all states communicate; reducible: not all states communicate.                                                                                          |
| Ergodic vs Non-ergodic chain                              | Ergodic: irreducible, aperiodic, positive recurrent; non-ergodic violates one of these.                                                                              |
| Limiting distribution                                     | Stationary distribution $\pi$ satisfying $\pi P = \pi$ .                                                                                                             |
| Gambler's Ruin vs Renewal Process                         | Ruin models wealth transitions; renewal process models independent event arrivals.                                                                                   |
| Periodicity vs Aperiodicity                               | Periodic: returns occur at multiples of $d > 1$ ; aperiodic: $d = 1$ .                                                                                               |

| Concept                        | Explanation                                            |
|--------------------------------|--------------------------------------------------------|
| Brownian Motion vs Random Walk | Brownian motion: continuous-time limit of random walk. |

## 2. Random Walk Mean and Variance

Let  $S_n = X_1 + X_2 + \dots + X_n$ , with  $X_i = \pm 1$  (p, q).

$$E[S_n] = n(2p - 1), \quad \text{Var}(S_n) = 4np(1 - p).$$

## 3. Markovian Property

$$P(X_{t+1} = j | X_t = i, X_{t-1}, \dots, X_0) = P(X_{t+1} = j | X_t = i).$$

The future depends only on the present.

## 4. Gambler's Ruin

Recurrence:  $P_i = pP_{i+1} + qP_{i-1}$ , with  $P_0 = 0, P_N = 1$ . - If  $p = q = 0.5$ :  $P_i = i/N$  - If  $p \neq q$ :  $P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}$

If  $p > q$ , probability of infinite gain =  $1 - q/p$ . If  $p \leq q$ , ruin is certain.

## 5. Components of Hidden Markov Model

1. Hidden states  $S$
2. Observations  $O$
3. Transition matrix  $A$
4. Emission matrix  $B$
5. Initial state distribution  $\pi$

## 6. Tom's Problem (p = 0.63, q = 0.37, N = 12)

$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}$  - Start  $i = 5$ :  $P_5 \approx 0.985$  - Infinitely rich:  $1 - q/p = 0.4127$  - Start  $i = 2$ :  $P_2 \approx 0.941 \Rightarrow P_{broke} = 0.059$

## 7. Collins' Stock Problem

$$p = 0.6, q = 0.4, i = 6, N = 21$$

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N} \approx 0.995.$$

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## 8. Patrick's Stock Problem

$$p = 0.55, q = 0.45, i = 5, N = 10$$

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N} \approx 0.874.$$

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## 9. HMM Assumptions

1. Hidden state process is Markovian.
  2. Each observation depends only on the current state.
  3. Model parameters (A, B,  $\pi$ ) are stationary.
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## 10. Pure Birth (Yule-Furry) Process Assumptions

1. Birth rate proportional to population:  $\lambda_n = n\lambda$ .
  2. No deaths.
  3. Independent births.
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## 11. Mean of Birth-Death Process

$$\frac{dm(t)}{dt} = (\lambda - \mu)m(t) \Rightarrow m(t) = m(0)e^{(\lambda - \mu)t}.$$

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## 12. Poisson Process Derivation

Using Chapman-Kolmogorov Postulates:

$$P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad E[N(t)] = \text{Var}[N(t)] = \lambda t.$$

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### 13. Markov Chain Graph (Generic)

- **Transition matrix:**  $P = [p_{ij}]$
  - **Communication class:** mutually reachable states.
  - **Ergodic:** irreducible + aperiodic.
  - **Limiting distribution:** exists for ergodic chains.
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### 14. Two-State Markov Chain Limit

$$P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \Rightarrow \lim_{n \rightarrow \infty} P^n = \frac{1}{a+b} \begin{bmatrix} b & a \\ b & a \end{bmatrix}.$$

Stationary distribution:  $\pi = (b/(a+b), a/(a+b))$ .

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### 15. M/M/1 Queue

$$\rho = \lambda/\mu < 1$$

$$L_s = \frac{\rho}{1-\rho}, \quad L_q = \frac{\rho^2}{1-\rho}, \quad W_s = \frac{1}{\mu-\lambda}, \quad W_q = \frac{\lambda}{\mu(\mu-\lambda)}.$$

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### 16. Components of a Queuing Model

1. Arrival process (e.g., Poisson).
  2. Service mechanism (distribution + servers).
  3. Queue discipline (FIFO, priority, etc.).
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### 17. Telephone Attendant Problem

$\lambda = 110/hr, \mu = 120/hr, \rho = 0.917$  - Utilization = 91.7% -  $W_q = \rho/(\mu - \lambda) = 0.0917 \text{ hr} \approx 5.5 \text{ min}$  -  $L_q = \lambda W_q = 10.1$  people in queue (avg)

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**End of Solutions**